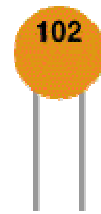


Capacitor Codes

A number code is often used on small capacitors where printing is difficult:

- the 1st number is the 1st digit,
- the 2nd number is the 2nd digit,
- the 3rd number is the number of zeros to give the capacitance in pF.
- Ignore any letters - they just indicate tolerance and voltage rating.

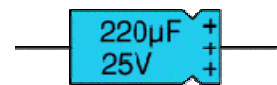


For example: **102** means $1000\text{pF} = 1\text{nF}$ (*not 102pF !*)

For example: **473J** means $47000\text{pF} = 47\text{nF} = 0.047\mu\text{F}$
(J means 5% tolerance).

Electrolytic Capacitors Polarised capacitors

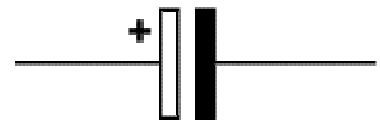
Axial Type >



Radial Type >



Circuit Symbol >



Electrolytic capacitors are polarised and they must be connected the correct way round, at least one of their leads will be marked + or -

There are two designs of electrolytic capacitors; axial where the leads are attached to each end ($220\mu\text{F}$ in picture) and radial where both leads are at the same end ($10\mu\text{F}$ in picture). Radial capacitors tend to be a little smaller and they stand upright on the circuit board.

It is easy to find the value of electrolytic capacitors because they are clearly printed with their capacitance and voltage rating.

Tantalum Bead Capacitors

The long leg denotes the + Positive which is also shown by a + to the nearest leg.

Tantalum bead capacitors are polarised and **they must be connected the correct way round**, they also have low voltage ratings like electrolytic capacitors. They are expensive but very small, so they are used where a large capacitance is needed in a small size.

Modern tantalum bead capacitors are printed with their capacitance and voltage in full eg

2u2 = $2.2\mu\text{F}$ and **u47 = $0.47\mu\text{F}$**

